

symlacerate_density

```
library(ggplot2)
library(ggpubr)
library(plotrix)
library(tidyverse)
library(dplyr)
library(car)
library(lme4)
library(emmeans)
library(qqplotr)
library(here)
library(patchwork)
library(multcomp)
library(multcompView)
library(broom)
library(rstatix)
library(janitor)
library(tibble)
library(magick)
library(grid)
```

```
rm(list = ls())
#graphics.off()
```

```
getwd()
```

```
## [1] "/Users/junbc/Documents/GitHub/heating_lacerates_final/code"
```

```
setwd("/Users/junbc/Documents/GitHub/heating_lacerates_final/")
```

```
# 1) Symbiotic lacerate symbiont density
```

```
symdensity_raw <- read.csv(
  "data/LacerateSymDensity.csv",
  check.names = FALSE
)
```

```
bad_names <- is.na(names(symdensity_raw)) | names(symdensity_raw) == ""
if (any(bad_names)) {
  names(symdensity_raw)[bad_names] <- paste0("V", seq_len(sum(bad_names)))
}
```

```
symdensity_filtered <- symdensity_raw %>%
  filter(
```

```

    Type == "Symbiont Area",
    day != "IGNORE",
    treatment != "IGNORE",
    Label != "Snap-1510.czi",
    Label != "Snap-1479.czi"
  ) %>%
  mutate(
    day = as.numeric(day),
    calculation = as.numeric(calculation)
  )

symdensity_wellmeans <- symdensity_filtered %>%
  group_by(day, treatment, well) %>%
  summarise(
    calculation = mean(calculation, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    day = factor(day, levels = c(0, 1, 5, 7, 9, 11, 13, 14)),
    treatment = factor(treatment, levels = c("Sym-Control", "Sym-HS"))
  )

# 2) Symbiotic lacerate statistics

symdensity_stats <- symdensity_wellmeans %>%
  mutate(
    day = factor(day, levels = c(0, 1, 5, 7, 9, 11, 13, 14)),
    treatment = factor(treatment, levels = c("Sym-Control", "Sym-HS"))
  )

mod_aov <- lm(
  calculation ~ treatment * day,
  data = symdensity_stats
)

anova_symdensity <- Anova(mod_aov, type = "II")
print(anova_symdensity)

```

```

## Anova Table (Type II tests)
##
## Response: calculation
##           Sum Sq Df F value    Pr(>F)
## treatment    4676.3   1 99.1439 < 2.2e-16 ***
## day          1150.9   7  3.4858  0.001291 **
## treatment:day  2566.5   7  7.7734 1.098e-08 ***
## Residuals    14951.8 317
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

anova_symdensity_df <- as.data.frame(anova_symdensity) %>%
  tibble::rownames_to_column("Factor")

```

```

write.csv(
  anova_syndensity_df,
  file = file.path(
    "~/Documents/GitHub/heating_lacerates_final/tables",
    "TableS4_anova_syndensity.csv"
  ),
  row.names = FALSE
)

emm_syndensity <- emmeans(mod_aov, ~ treatment | day)
tukey_syndensity <- summary(pairs(emm_syndensity, adjust = "tukey"))
print(tukey_syndensity)

```

```

## day = 0:
## contrast          estimate    SE  df t.ratio p.value
## (Sym-Control) - (Sym-HS)   -1.46 1.98 317  -0.735  0.4631
##
## day = 1:
## contrast          estimate    SE  df t.ratio p.value
## (Sym-Control) - (Sym-HS)    4.17 1.98 317   2.103  0.0362
##
## day = 5:
## contrast          estimate    SE  df t.ratio p.value
## (Sym-Control) - (Sym-HS)    7.88 2.11 317   3.735  0.0002
##
## day = 7:
## contrast          estimate    SE  df t.ratio p.value
## (Sym-Control) - (Sym-HS)    7.79 2.22 317   3.516  0.0005
##
## day = 9:
## contrast          estimate    SE  df t.ratio p.value
## (Sym-Control) - (Sym-HS)    6.86 2.14 317   3.205  0.0015
##
## day = 11:
## contrast          estimate    SE  df t.ratio p.value
## (Sym-Control) - (Sym-HS)    6.87 2.20 317   3.126  0.0019
##
## day = 13:
## contrast          estimate    SE  df t.ratio p.value
## (Sym-Control) - (Sym-HS)   15.21 2.21 317   6.896 <0.0001
##
## day = 14:
## contrast          estimate    SE  df t.ratio p.value
## (Sym-Control) - (Sym-HS)   18.09 2.45 317   7.395 <0.0001

```

```

tukey_syndensity_df <- as.data.frame(tukey_syndensity)

write.csv(
  tukey_syndensity_df,
  file = file.path(
    "~/Documents/GitHub/heating_lacerates_final/tables",
    "TableS5_tukey_syndensity.csv"
  ),

```

```

    row.names = FALSE
  )

tukey_df <- as.data.frame(tukey_syndensity)

letter_df <- tukey_df %>%
  mutate(
    group_control = "a",
    group_hs = ifelse(p.value < 0.05, "b", "a")
  ) %>%
  select(day, group_control, group_hs)

letter_df_long <- letter_df %>%
  pivot_longer(
    cols = c(group_control, group_hs),
    names_to = "group",
    values_to = ".group"
  ) %>%
  mutate(
    treatment = ifelse(group == "group_control", "Sym-Control", "Sym-HS")
  ) %>%
  select(day, treatment, .group)

letter_positions <- syndensity_stats %>%
  group_by(day, treatment) %>%
  summarise(
    y_pos = max(calculation, na.rm = TRUE) + 2.9,
    .groups = "drop"
  ) %>%
  left_join(letter_df_long, by = c("day", "treatment"))

# 4) Final symbiotic lacerate plot

theme_set(
  theme_bw(base_size = 12, base_family = "sans")
)

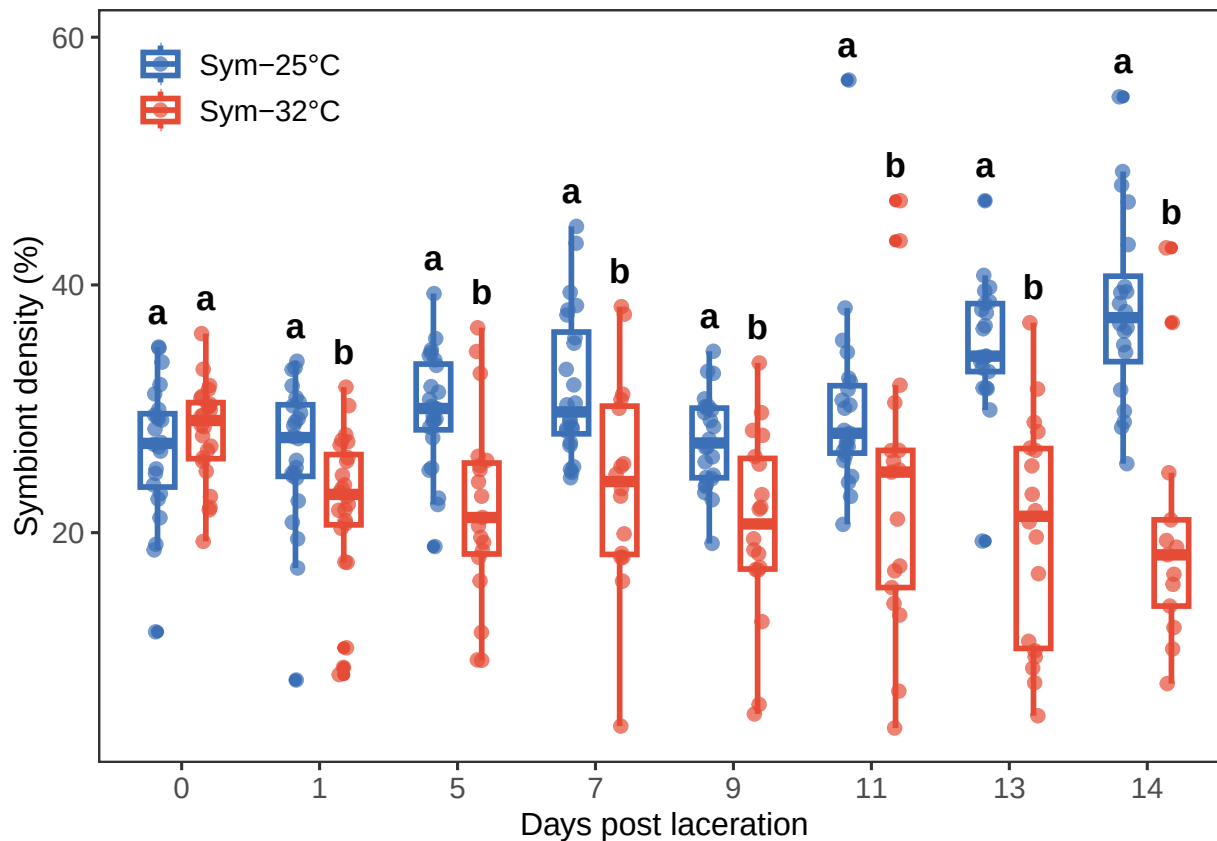
p_syndensity_final <- ggplot(
  syndensity_stats,
  aes(x = day, y = calculation, color = treatment)
) +
  geom_jitter(
    position = position_jitterdodge(jitter.width = 0.15, dodge.width = 0.7),
    size = 2,
    alpha = 0.7
  ) +
  geom_boxplot(
    aes(group = interaction(day, treatment)),
    position = position_dodge(width = 0.7),
    width = 0.5,
    fill = NA,

```

```

    linewidth = 1
  ) +
  geom_text(
    data = letter_positions,
    aes(x = day, y = y_pos, label = .group, group = treatment),
    position = position_dodge(width = 0.7),
    inherit.aes = FALSE,
    size = 4.5,
    fontface = "bold",
    family = "sans"
  ) +
  scale_color_manual(
    values = c("Sym-Control" = "#3B6FB6", "Sym-HS" = "#E64B35"),
    labels = c("Sym-25°C", "Sym-32°C")
  ) +
  labs(
    x = "Days post laceration",
    y = "Symbiont density (%)"
  ) +
  theme(
    panel.grid = element_blank(),
    legend.title = element_blank(),
    axis.title = element_text(size = 12),
    axis.text = element_text(size = 11),
    legend.text = element_text(size = 11),
    legend.position = c(0.02, 0.98),
    legend.justification = c(0, 1),
    legend.background = element_blank(),
    legend.key = element_blank()
  )
p_syndensity_final

```



6) Inoculated lacerate symbiont density

```
inoc_raw <- read.csv(
  "data/Lacerate-Inoc-Sym-Density.csv",
  check.names = FALSE
)

bad_names <- is.na(names(inoc_raw)) | names(inoc_raw) == ""
if (any(bad_names)) {
  names(inoc_raw)[bad_names] <- paste0("V", seq_len(sum(bad_names)))
}

inoc_filtered <- inoc_raw %>%
  filter(Type == "Symbiont Area") %>%
  filter(is.na(notes) | notes == "") %>%
  mutate(
    day = as.numeric(day),
    calculation = as.numeric(calculation),
    treatment = as.character(treatment)
  ) %>%
  filter(
    treatment %in% c("inoc-25C", "inoc-32C"),
    !is.na(day),
    !is.na(calculation)
  )
```

```

inoc_wellmeans <- inoc_filtered %>%
  group_by(day, treatment, well) %>%
  summarise(
    calculation = mean(calculation, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    day = factor(day, levels = sort(unique(day))),
    treatment = factor(treatment, levels = c("inoc-25C", "inoc-32C"))
  )

```

7) Inoculated lacerate statistics

```

mod_inoc <- lm(
  calculation ~ treatment * day,
  data = inoc_wellmeans
)

```

```

anova_inoc <- Anova(mod_inoc, type = "II")
print(anova_inoc)

```

Anova Table (Type II tests)

##

Response: calculation

##		Sum Sq	Df	F value	Pr(>F)
## treatment		26471.1	1	319.4869	< 2.2e-16 ***
## day		2330.0	7	4.0173	0.0003383 ***
## treatment:day		1009.3	7	1.7403	0.0998962 .
## Residuals		21708.0	262		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

anova_inoc_df <- as.data.frame(anova_inoc) %>%
  tibble::rownames_to_column("Factor")

write.csv(
  anova_inoc_df,
  file = file.path(
    "~/Documents/GitHub/heating_lacerates_final/tables",
    "TableS6_anova_inoc.csv"
  ),
  row.names = FALSE
)

```

```

emm_inoc <- emmeans(mod_inoc, ~ treatment | day)
tukey_inoc <- summary(pairs(emm_inoc, adjust = "tukey"))
print(tukey_inoc)

```

day = 7:

## contrast	estimate	SE	df	t.ratio	p.value
## (inoc-25C) - (inoc-32C)	14.3	3.60	262	3.977	<0.0001

```
##
## day = 10:
## contrast estimate SE df t.ratio p.value
## (inoc-25C) - (inoc-32C) 14.6 2.97 262 4.905 <0.0001
##
## day = 12:
## contrast estimate SE df t.ratio p.value
## (inoc-25C) - (inoc-32C) 18.5 2.80 262 6.614 <0.0001
##
## day = 14:
## contrast estimate SE df t.ratio p.value
## (inoc-25C) - (inoc-32C) 22.0 2.96 262 7.408 <0.0001
##
## day = 16:
## contrast estimate SE df t.ratio p.value
## (inoc-25C) - (inoc-32C) 23.6 2.92 262 8.055 <0.0001
##
## day = 18:
## contrast estimate SE df t.ratio p.value
## (inoc-25C) - (inoc-32C) 25.7 2.92 262 8.799 <0.0001
##
## day = 20:
## contrast estimate SE df t.ratio p.value
## (inoc-25C) - (inoc-32C) 22.1 3.65 262 6.057 <0.0001
##
## day = 21:
## contrast estimate SE df t.ratio p.value
## (inoc-25C) - (inoc-32C) 21.5 5.70 262 3.766 0.0002
```

```
tukey_inoc_df <- as.data.frame(tukey_inoc)
```

```
write.csv(
  tukey_inoc_df,
  file = file.path(
    "~/Documents/GitHub/heating_lacerates_final/tables",
    "TableS7_tukey_inoc.csv"
  ),
  row.names = FALSE
)
```

```
tukey_inoc_df <- as.data.frame(tukey_inoc)
```

```
letter_df <- tukey_inoc_df %>%
  mutate(
    group_25 = "a",
    group_32 = ifelse(p.value < 0.05, "b", "a")
  ) %>%
  select(day, group_25, group_32)
```

```
letter_df_long <- letter_df %>%
  pivot_longer(
    cols = c(group_25, group_32),
    names_to = "group",
```

```

    values_to = ".group"
  ) %>%
  mutate(
    treatment = ifelse(group == "group_25", "inoc-25C", "inoc-32C")
  ) %>%
  select(day, treatment, .group)

letter_positions <- inoc_wellmeans %>%
  group_by(day, treatment) %>%
  summarise(
    y_pos = max(calculation, na.rm = TRUE) * 1.12,
    .groups = "drop"
  ) %>%
  left_join(letter_df_long, by = c("day", "treatment"))

# 9) Final inoculated lacerate plot

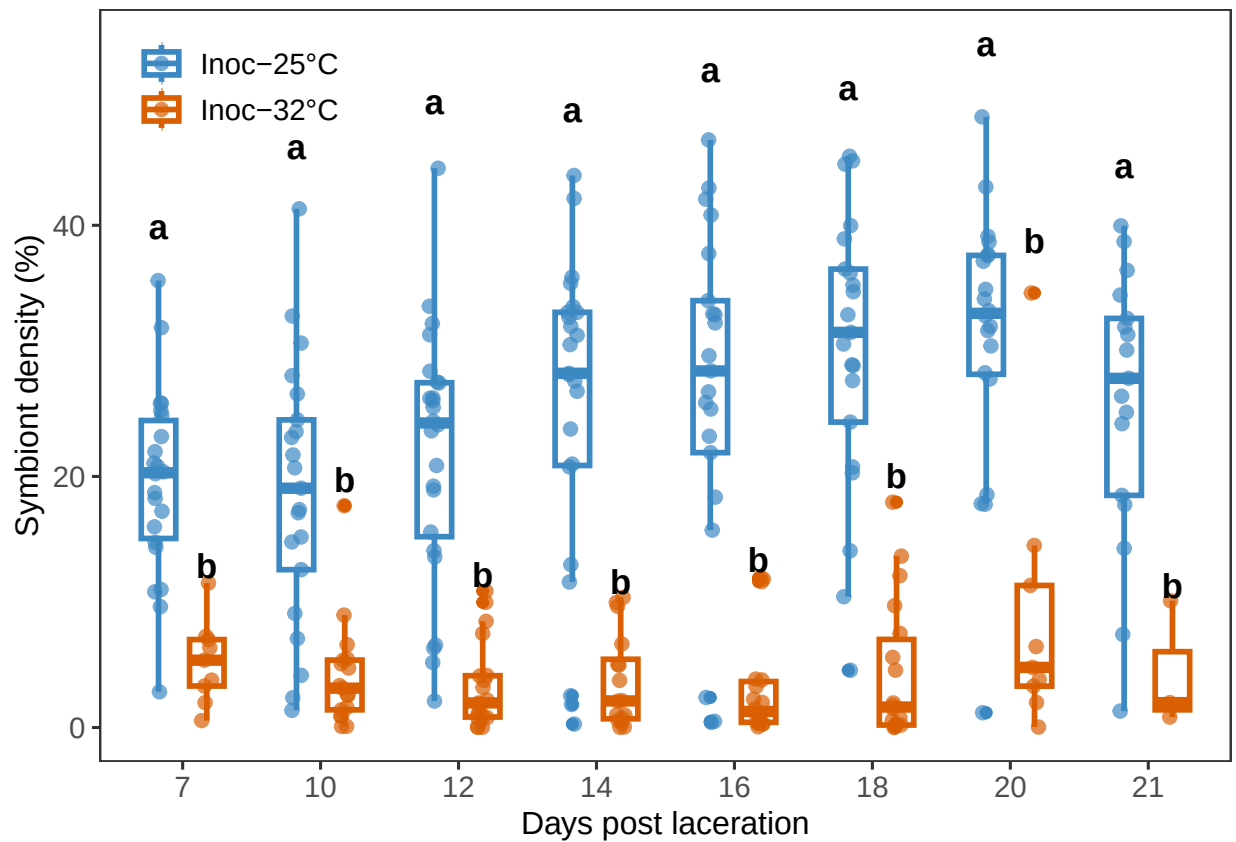
p_inoc_syndensity <- ggplot(
  inoc_wellmeans,
  aes(x = day, y = calculation, color = treatment)
) +
  geom_jitter(
    position = position_jitterdodge(jitter.width = 0.15, dodge.width = 0.7),
    size = 2,
    alpha = 0.7
  ) +
  geom_boxplot(
    aes(group = interaction(day, treatment)),
    position = position_dodge(width = 0.7),
    width = 0.5,
    fill = NA,
    linewidth = 1
  ) +
  geom_text(
    data = letter_positions,
    aes(x = day, y = y_pos, label = .group, group = treatment),
    position = position_dodge(width = 0.7),
    inherit.aes = FALSE,
    size = 4.5,
    fontface = "bold"
  ) +
  scale_color_manual(
    values = c("inoc-25C" = "#3B88C3", "inoc-32C" = "#D95F02"),
    labels = c("Inoc-25°C", "Inoc-32°C")
  ) +
  labs(
    x = "Days post laceration",
    y = "Symbiont density (%)"
  ) +
  theme_bw(base_size = 12) +
  theme(
    panel.grid = element_blank(),

```

```

legend.title = element_blank(),
axis.title = element_text(size = 12),
axis.text = element_text(size = 11),
legend.text = element_text(size = 11),
legend.position = c(0.02, 0.98),
legend.justification = c(0, 1),
legend.background = element_blank(),
legend.key = element_blank()
)
p_inoc_syndensity

```



```

img_meta_c8 <- tribble(
  ~row_lab,      ~col_lab, ~file,

  "Sym\n25°C",  "BF", "~Documents/Github/heating_lacerates_final/images/panel_inoc/S25_dp14_bf.png",
  "Sym\n25°C",  "FL", "~Documents/Github/heating_lacerates_final/images/panel_inoc/S25_dp14_fl.png",

  "Sym\n32°C",  "BF", "~Documents/Github/heating_lacerates_final/images/panel_inoc/S32_dp14_bf.png",
  "Sym\n32°C",  "FL", "~Documents/Github/heating_lacerates_final/images/panel_inoc/S32_dp14_fl.png",

  "Inoc\n25°C", "BF", "~Documents/Github/heating_lacerates_final/images/panel_inoc/I25_dp14_bf.png",
  "Inoc\n25°C", "FL", "~Documents/Github/heating_lacerates_final/images/panel_inoc/I25_dp14_fl.png",

  "Inoc\n32°C", "BF", "~Documents/Github/heating_lacerates_final/images/panel_inoc/I32_dp14_bf_Snap-9

```

```

    "Inoc\n32°C", "FL", "~/Documents/Github/heating_lacerates_final/images/panel_inoc/I32_dpl14_fl_Snap-9
  )

img_meta_c8

```

```

## # A tibble: 8 x 3
##   row_lab      col_lab file
##   <chr>        <chr>  <chr>
## 1 "Sym\n25°C" BF      ~/Documents/Github/heating_lacerates_final/images/panel_~
## 2 "Sym\n25°C" FL      ~/Documents/Github/heating_lacerates_final/images/panel_~
## 3 "Sym\n32°C" BF      ~/Documents/Github/heating_lacerates_final/images/panel_~
## 4 "Sym\n32°C" FL      ~/Documents/Github/heating_lacerates_final/images/panel_~
## 5 "Inoc\n25°C" BF      ~/Documents/Github/heating_lacerates_final/images/panel_~
## 6 "Inoc\n25°C" FL      ~/Documents/Github/heating_lacerates_final/images/panel_~
## 7 "Inoc\n32°C" BF      ~/Documents/Github/heating_lacerates_final/images/panel_~
## 8 "Inoc\n32°C" FL      ~/Documents/Github/heating_lacerates_final/images/panel_~

```

```

make_image_panel_c8 <- function(img_meta_c8) {

  row_levels <- unique(img_meta_c8$row_lab)
  col_levels <- unique(img_meta_c8$col_lab)

  img_meta_c8 <- img_meta_c8 %>%
    mutate(
      row_lab = factor(row_lab, levels = rev(row_levels)),
      col_lab = factor(col_lab, levels = col_levels),
      row_num = as.numeric(row_lab),
      col_num = as.numeric(col_lab)
    )

  image_size <- 0.98
  col_gap <- 1.04
  row_gap <- 1.12

  img_meta_c8 <- img_meta_c8 %>%
    mutate(
      x_center = col_num * col_gap + 0.22,
      y_center = row_num * row_gap
    )

  p <- ggplot() +
    xlim(0.55, 2.88) +
    ylim(0.35, max(img_meta_c8$y_center) + 0.65) +
    theme_void(base_family = "sans")

  for (i in seq_len(nrow(img_meta_c8))) {
    if (file.exists(img_meta_c8$file[i])) {
      img <- image_read(img_meta_c8$file[i])
      grob <- rasterGrob(as.raster(img), interpolate = TRUE)

      p <- p + annotation_custom(
        grob,
        xmin = img_meta_c8$x_center[i] - image_size / 2,

```

```

      xmax = img_meta_c8$x_center[i] + image_size / 2,
      ymin = img_meta_c8$y_center[i] - image_size / 2,
      ymax = img_meta_c8$y_center[i] + image_size / 2
    )
  }
}

row_df <- img_meta_c8 %>%
  distinct(row_lab, y_center) %>%
  mutate(lab = as.character(row_lab))

p +
  geom_text(
    data = row_df,
    aes(x = 0.80, y = y_center, label = lab),
    hjust = 1,
    size = 3.8,
    fontface = "bold",
    family = "sans"
  ) +
  annotate(
    "text",
    x = 1.24,
    y = 0.35,
    label = "BF",
    size = 3.8,
    fontface = "bold",
    family = "sans"
  ) +
  annotate(
    "text",
    x = 2.28,
    y = 0.35,
    label = "FL",
    size = 3.8,
    fontface = "bold",
    family = "sans"
  )
}

image_panel_c8 <- make_image_panel_c8(img_meta_c8)

matched_legend_theme <- theme(
  legend.position = c(0.03, 0.97),
  legend.justification = c(0, 1),
  legend.background = element_blank(),
  legend.key = element_blank(),
  legend.title = element_blank(),
  legend.text = element_text(size = 11),
  legend.margin = margin(0, 0, 0, 0),
  legend.box.margin = margin(0, 0, 0, 0)
)

```

```

p_syndensity_final <- p_syndensity_final +
  coord_cartesian(ylim = c(0, 60)) +
  matched_legend_theme +
  theme(
    plot.margin = margin(t = 5, r = 5, b = 2, l = 5),
    axis.title.x = element_text(margin = margin(t = 1))
  )

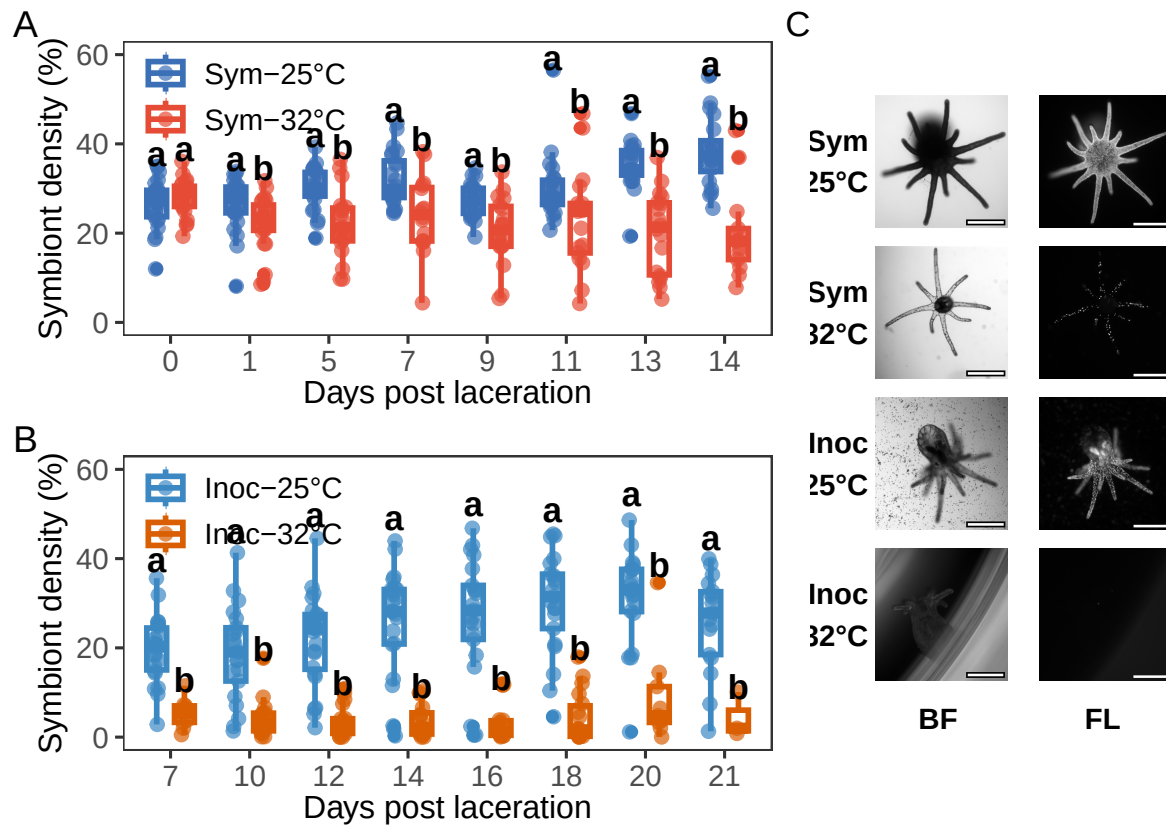
p_inoc_syndensity <- p_inoc_syndensity +
  coord_cartesian(ylim = c(0, 60)) +
  matched_legend_theme +
  theme(
    plot.margin = margin(t = 2, r = 5, b = 0, l = 5),
    axis.title.x = element_text(margin = margin(t = 1))
  )

left_panels <- p_syndensity_final / p_inoc_syndensity +
  plot_layout(heights = c(1, 1))

final_fig4 <- wrap_plots(
  left_panels,
  image_panel_c8,
  ncol = 2,
  widths = c(1.60, 1.00)
) +
  plot_annotation(
    tag_levels = "A",
    theme = theme(
      text = element_text(family = "sans"),
      plot.tag = element_text(size = 16, face = "bold"),
      plot.tag.position = c(0.01, 0.99)
    )
  )

print(final_fig4)

```



```
ggsave(
  filename = "Fig4.png",
  plot = final_fig4,
  path = "~/Documents/Github/heating_lacerates_final/figs/",
  device = "png",
  # width = 7.2,
  # height = 4.2,
  width = 10,
  height = 6.75,
  units = "in",
  dpi = 600,
  ## compression "lzw",
  bg = "white"
)

ggsave(
  filename = "Fig4.pdf",
  plot = final_fig4,
  path = "~/Documents/Github/heating_lacerates_final/figs/",
  device = pdf,
  # width = 7.2,
  # height = 4.2,
  width = 15,
  height = 6.5,
  units = "in",
  bg = "white"
)
```

)